

MapAnalysisDemo — Technical Specification

Version: 2.0
Last Updated: May 2026
Type: Single-Page Analytics Dashboard with Map Visualization

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1. Project Overview

MapAnalysisDemo is a self-contained, single-file analytics dashboard with interactive map visualization. Built entirely in `index.html` (~2243 lines), it uses **vanilla JavaScript**, **Leaflet.js** for mapping, and **SheetJS** for spreadsheet parsing — no build tools, bundlers, or frameworks.

Primary Use Case

Import CSV/XLSX files with geographic coordinates, visualize them on an interactive map, explore data in a sortable/filterable table, analyze with a distance measurement tool, and customize presentation (icons, labels, colors) per-dataset and per-point.

Key Design Principles

Principle	Description
Single-file	Entire application in one HTML file — zero deployment friction
No build step	Vanilla JS with CDN-loaded libraries
Immediate startup	Ships with 3 sample datasets loaded on page load
Self-contained data	4 built-in POI datasets generated synthetically (no API calls)
Stateful modals	Each operation uses a consistent modal pattern with target state variables

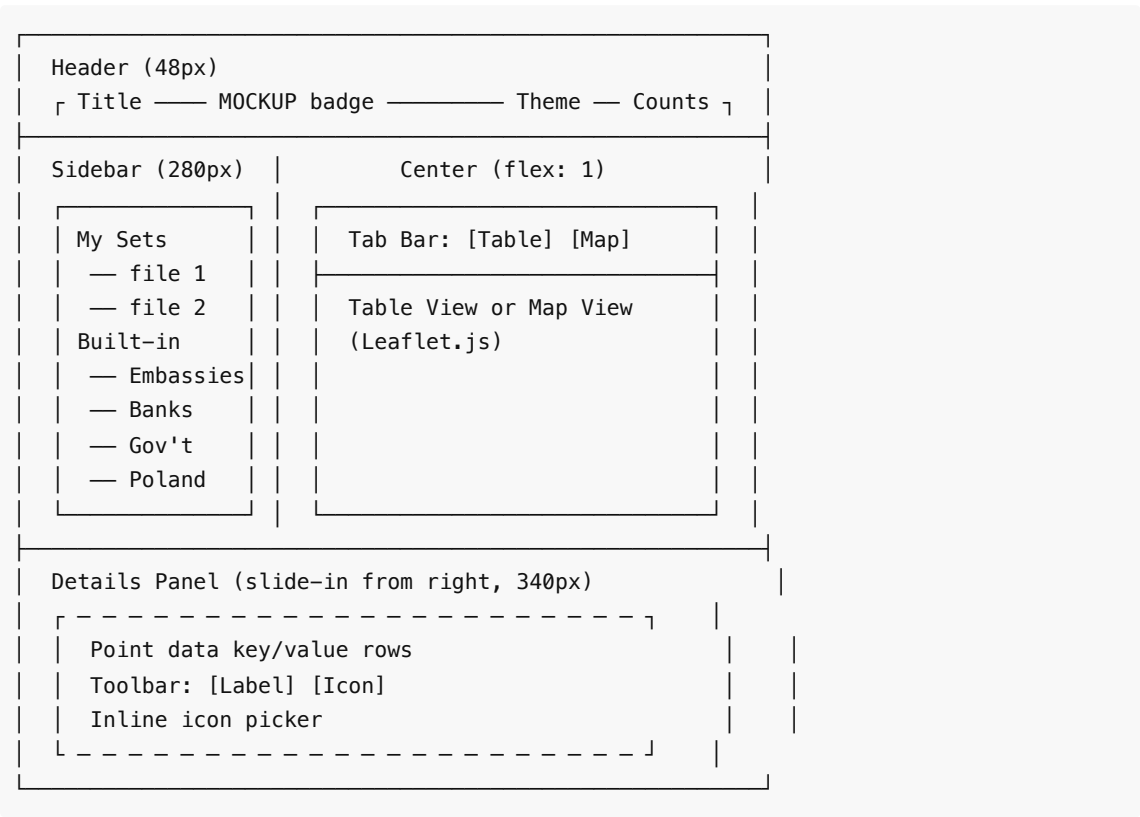
External Dependencies

Library	Version	Source	Purpose
Leaflet.js	1.9.4	unpkg.com/leaflet@1.9.4	Interactive map rendering
SheetJS	0.20.3	cdn.sheetjs.com/xlsx-0.20.3	Excel/CSV file parsing

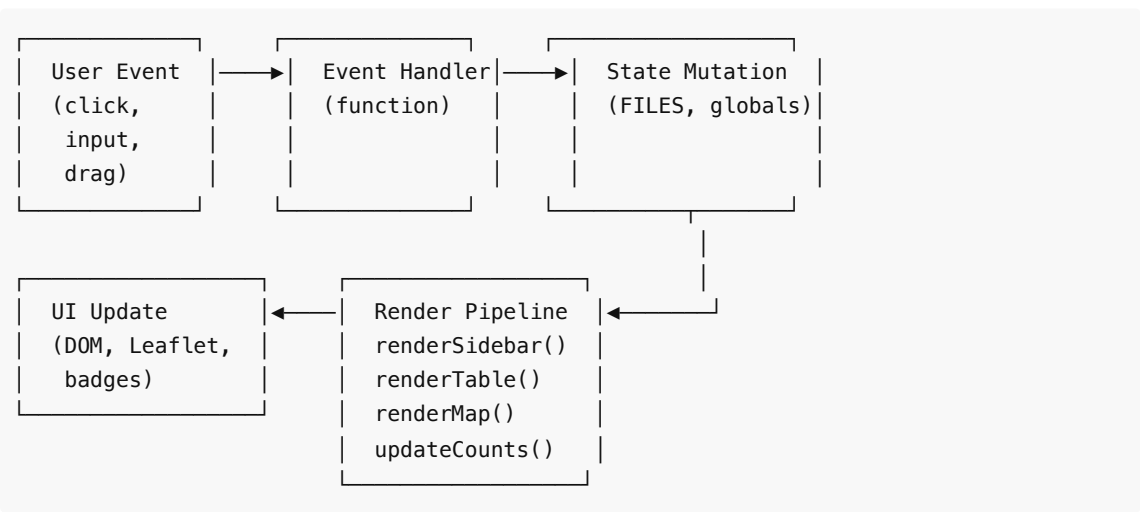
CartoDB	—	basemaps.cartocdn.com	Map tile layers (dark + light)
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2. Architecture

2.1 Layout Structure



2.2 Data Flow



2.3 Module Organization (within single file)

Section	Lines (approx.)	Content
CSS	1–380	Custom properties, reset, layout, sidebar, table, map, details panel, modals, themes, animations
HTML	381–680	Static DOM structure: header, sidebar, main panes, details panel, modal dialogs
JS - Data	681–900	Built-in set definitions, icon picker options, sample data generators
JS - State	901–950	Global variables and initialization
JS - Core	951–1400	Rendering functions, event handlers, dialog logic
JS - Map	1401–1700	Leaflet integration, measurement tool, blink animation
JS - Utility	1701–1800	<code>parseCoord()</code> , <code>detectCoordFormat()</code> , <code>formatDist()</code> , <code>getLabelText()</code>
JS - Init	1801–end	Start-up sequence, initial render calls

3. Feature Catalog

3.1 Data Management

ID	Feature	Description	Trigger	Status
F1	Import CSV/XLSX	Open file dialog for .xlsx, .xls, .csv files. Parses with SheetJS, shows preview with column mapping	+ Import button	✓
F2	Coordinate auto-detection	Automatically identifies lat/lng columns by regex <code>(/lat/i, /lng lon/i)</code>	On import dialog open	✓
F3	Multi-format coordinate parsing	Parses DD (decimal), DDM (deg min), DMS (deg min sec) with directional suffixes (N/S/E/W)	On confirm import	✓
F4	Merge datasets	Union-merge selected files' data — collects all unique keys, fills missing values with ''	Merge button	✓
F5	Remove dataset	Confirmation dialog; re-selects nearest neighbor if active file removed	× on file item	✓
F6	Rename dataset	Inline text input with auto-focus + select-all; Enter key to confirm	🖋 on file item	✓
F7	Dataset visibility	Show/hide dataset on map; visual red strikethrough on eye icon when hidden	Eye button on file item	✓
F8	Built-in POI sets	4 synthetic datasets: Embassies in Europe, Banks in Europe, Government Locations,	"Built-in" sidebar tab	✓

		Embassies in Poland		
F9	Add new point	Modal with dynamic fields matching active dataset schema + map-click coordinate picking	+ Point button	→ SOON
F10	Dataset-level icon	20-emoji picker applied to all points in a dataset	🌈 on file item	✓

3.2 Map Visualization

ID	Feature	Description	Status
M1	Point markers	Circle markers (color-coded per dataset, radius 7, white stroke) or emoji divIcon overlays	✓
M2	Per-point icon override	Inline emoji picker in details panel sets <code>row._icon</code> for an individual point	✓
M3	Track mode	Polyline connecting points in row order; start marker (green) + end marker (red) with popups	✓
M4	Combined mode	Both point markers and track polyline visible simultaneously	✓
M5	Dataset labels	Toggle labels for all points in a dataset via <code>file.showLabels</code>	✓
M6	Per-point labels	Toggle label for individual point via <code>row._showLabel</code> (overrides dataset setting)	✓
M7	Auto label extraction	<code>getLabelText()</code> checks name, city, store, respondent keys; falls back to <code>#id</code> or first non-null value	✓
M8	Per-point custom label	Text input in details panel to set arbitrary label text via <code>row._label</code>	→ SOON
M9	Map click → details	Click any marker to open the details panel with all row data	✓
M10	Zoom to dataset	Clicking a sidebar dataset fits map to its bounding box	✓
M11	Blink animation	Selected dataset markers flash red → fade back to original over 2s (1s each direction)	✓
M12	Measurement tool	Click map to place draggable waypoints; dashed polylines with segment distances + cumulative total	✓
M13	Theme-aware tiles	Dark CartoDB tiles in dark mode, light tiles in light mode, swapped via <code>tileLayer.setUrl()</code>	✓
M14	Map click coordinate pick	Map enters crosshair mode, single click fills both lat + lng fields (for Add Point)	→ SOON
M15	Popup on click	Shows all non-lat/lng fields as <code>key: value</code> pairs	✓

3.3 Table View

ID	Feature	Description	Status
T1	Column sorting	Click column header to sort ascending/descending; visual arrow indicator (▲/▼)	✓
T2	Global text search	Free-text filter across all columns simultaneously	✓
T3	Per-column filters	Individual text input under each column header for column-specific filtering	✓
T4	Row selection → details	Click any row to open details panel and center map on that point (zoom 10)	✓
T5	Row count display	Shows "X of Y rows" with filtered count / total count	✓
T6	Clear filter button	x button appears when filters are active, clears global search	✓

3.4 Details Panel

ID	Feature	Description	Status
D1	Slide-in panel	Animated 340px panel slides from right; 0.3s cubic-bezier transition	✓
D2	Key/value display	All row fields rendered as styled key/value rows with bottom borders	✓
D3	Per-point label toggle	Override dataset-level label visibility for individual point	✓
D4	Per-point icon picker	Inline 6-column emoji grid; sets row._icon override	✓
D5	Per-point label editing	Text input field to set custom label text via row._label	→ SOON

3.5 Theme & UX

ID	Feature	Description	Status
U1	Dark/Light theme	Moon toggle in header swaps all CSS custom properties and map tiles	✓
U2	Main tab switching	Table / Map tabs with active state and count badges	✓
U3	Total record counter	Header displays sum of file.rows across all datasets	✓
U4	Sidebar tab system	"My Sets" (imported) and "Built-in" (synthetic) views with active state	✓
U5	Loading indicators	status = 'loading' disables built-in POI items during data generation	✓
U6	Empty states	Placeholder content when no datasets exist or import dialog is empty	✓

U7	Measurement distance	Gold-styled distance markers at segment midpoints; "Total: X.XX km"	✓
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4. Function Catalog

4.1 Data Generation Functions

Function	Signature	Returns	Description
generateSalesData	() → Array<Object>	15 rows	City sales data: id, city, lat, lng, region, revenue, units, growth
generateStoreData	() → Array<Object>	12 rows	NYC store data: id, store, lat, lng, type, sqft, employees
generateSurveyData	() → Array<Object>	20 rows	Survey responses: id, respondent, lat, lng, region, satisfaction, age
generateEmbassiesData	() → Array<Object>	36–126 rows	European embassies across 18 capitals; 2–6 per city, jittered $\pm 0.02^\circ$
generateBanksData	() → Array<Object>	30–70 rows	European banks across 10 financial hubs; 3–7 per city, deduped, jittered $\pm 0.015^\circ$
generateGovernmentData	() → Array<Object>	36–84 rows	European govt buildings across 12 capitals; 3–6 per city, deduped, jittered $\pm 0.0125^\circ$
generatePolandEmbassiesData	() → Array<Object>	70 rows	Embassies in Warsaw; 70 countries positioned radially around city center

4.2 Rendering Functions

Function	Signature	Description
renderBuiltinSets	() → void	Renders BUILTIN_SETS array into #builtinList DOM with icons, names, and status text
renderSidebar	() → void	Renders FILES array into #fileList DOM with per-file action buttons (visibility, labels, icon, rename, remove)
renderTable	() → void	Orchestrates full table render: calls renderTableHead() then renderTableBody()
renderTableHead	() → void	Builds <thead> with sortable column headers (sort arrows + click handlers) and per-column filter inputs
renderTableBody	() → void	Filters via getFilteredData(), sorts via sortData(), renders <tbody> rows; updates #currentFileLabel, #rowInfo, #clearFilter visibility

renderMap	(focusFileId?: string) → void	Full map re-render: clears all non-tile layers, iterates visible files, draws points/tracks, fit-bounds to focus file or all visible coords
renderMeasurements	() → void	Re-renders measure tool: draggable gold circle markers, dashed polylines, midpoint distance labels, total distance display
renderImportPreview	() → void	Renders first 5 rows of pendingImport into the import dialog's preview table

4.3 Event Handlers / UI Callbacks

Function	Signature	Description
switchSidebarTab	(tab: 'my' 'builtin') → void	Toggles between My Sets and Built-in sidebar views
selectFile	(id: string) → void	Sets file active, resets sort, re-renders everything, triggers blink animation
blinkDataset	(id: string) → void	Animates dataset markers: red flash (sepia + blur filter) for 1s, restored over 1s
toggleVisibility	(id: string) → void	Toggles file.visible, re-renders sidebar (strikethrough eye) and map
toggleDatasetLabels	(id: string) → void	Toggles file.showLabels, clears all per-point _showLabel overrides, re-renders
getLabelText	(row: object) → string	Extracts display label: checks name, city, store, respondent, _label, then #id, then first non-null value
showIconPicker	(id: string) → void	Opens dataset icon picker modal with 20 emoji options
selectIcon	(ic: string null) → void	Sets file.icon, closes dialog, re-renders sidebar + map
cancelIconPicker	() → void	Clears iconPickerTargetId, closes modal
selectTableRow	(idx: number) → void	Selects row, opens details panel, centers map on point at zoom 10
setMapMode	(mode: 'points' 'track' 'both') → void	Updates toggle buttons, re-renders map in selected mode
switchTab	(name: 'table' 'map') → void	Switches main content tab; invalidates map size on map tab switch
updateCounts	() → void	Updates tab badges (#tableCount, #mapCount) and header #recordCount (sum of all file rows)

toggleTheme	() → void	Toggles .light class on <body>, swaps map tile URL via tileLayer.setUrl()
importFile	() → void	Triggers click on hidden <input #fileInput> to open OS file dialog
handleFileImport	(event: InputEvent) → void	Reads file via FileReader, parses with SheetJS (array buffer for xlsx, text for csv), stores in pendingImport, calls showImportDialog()
showImportDialog	() → void	Populates lat/lng column selects with auto-detection + coordinate format detection, shows preview
confirmImport	() → void	Creates new FILES entry from pendingImport, parses coordinates via parseCoord(), re-renders everything
cancelImport	() → void	Clears pendingImport, closes import dialog
showMergeDialog	() → void	Opens merge dialog with checkboxes for all existing files (active file pre-checked)
confirmMerge	() → void	Validates ≥2 files selected, union-merges data (all keys, missing values filled with ''), creates new file entry, re-renders
cancelMerge	() → void	Closes merge dialog
showRemoveDialog	(id: string) → void	Opens remove confirmation with file name displayed
confirmRemove	() → void	Removes file from FILES, selects nearest neighbor if active removed, re-renders
cancelRemove	() → void	Clears removeTargetId, closes dialog
showRenameDialog	(id: string) → void	Opens rename dialog pre-filled with current name, auto-focuses + selects all text
confirmRename	() → void	Validates non-empty name, updates file.name, re-renders sidebar + table
cancelRename	() → void	Clears renameTargetId, closes dialog
onFilterChange	() → void	Re-renders table body and map (for filtered data) when any filter input changes
clearGlobalFilter	() → void	Clears #globalFilter value, calls onFilterChange()
sortBy	(key: string) → void	Toggles sort direction if same key, or sets new sort key ascending; updates sortKey/sortDir globals, re-renders
toggleMeasure	() → void	Toggles measurement mode: attaches/removes map click handler, changes cursor to crosshair

addMeasurePoint	(e: LeafletMouseEvent) → void	Pushes {lat, lng} to measurePoints, shows clear button, calls renderMeasurements()
clearMeasurements	() → void	Clears measurePoints, removes measure layers, hides measure UI
loadBuiltinSet	(id: string) → void	Sets status to 'loading', calls appropriate generator, creates file entry, deactivates others, switches to My Sets tab, re-renders
openDetails	(row: object) → void	Opens details panel: sets title, renders toolbar (Label/Icon toggles), inline icon picker, all key/value rows
togglePointIcon	() → void	Shows/hides #pointIconPicker inline grid in details panel
selectPointIcon	(ic: string null) → void	Sets row._icon, re-opens details panel, re-renders map
togglePointLabel	() → void	Toggles row._showLabel boolean, re-opens details panel, re-renders map
closeDetails	() → void	Closes details panel, clears currentDetailRow
showAddPointDialog	() → void	(Planned) Opens Add Point modal with dynamic fields from active dataset schema
enableMapPick	() → void	(Planned) Puts map in crosshair selection mode for coordinate picking
onMapPick	(e: LeafletMouseEvent) → void	(Planned) Fills lat + lng fields after map click in pick mode
confirmAddPoint	() → void	(Planned) Appends new row to file.data, re-renders everything

4.4 Utility Functions

Function	Signature	Returns	Description
getCurrentFile	() → FileEntry undefined	Active FILES entry matching currentFileId	
getLatLngColumns	(file: FileEntry) → {latKey: string, lngKey: string}	Identifies lat/lng column names: checks saved keys first, falls back to regex detection	
getFilteredData	() → Array<Object>	Applies global filter + per-column filters to current file's data; returns filtered array	

sortData	(arr: Array<Object>) → Array<Object>	Returns sorted shallow copy based on sortKey/sortDir globals; safe for non-existent keys	
formatDist	(m: number) → string	Formats distance: ≥1000m → "X.XX km", <1000m → "X.X m"	
detectCoordFormat	(values: string[]) → 'DD' 'DDM' 'DMS'	Heuristic: checks for ° with " → DMS, ° with ' → DDM, else DD	
parseCoord	(value: string number, format: 'DD' 'DDM' 'DMS') → number NaN	Parses coordinate string to decimal degrees; handles directional suffixes (N/S/E/W), trims whitespace	

5. Data Structures & State

5.1 FILES — Master Dataset Registry (Global Array)

```
interface FileEntry {
  id: string;           // Auto-incrementing: 'f1', 'f2', ...
  name: string;         // Display name (original filename or user-renamed)
  type: 'xlsx' | 'csv'; // Source file type
  size: string;         // Human-readable: '2.4 MB', '47 points'
  rows: number;        // Number of data rows
  cols: number;        // Number of columns
  active: boolean;      // Currently selected/highlighted in sidebar
  visible: boolean;     // Shown on map
  icon: string | null;  // Dataset-level emoji override (from picker)
  showLabels: boolean;  // Show labels on all points
  latKey: string;       // Latitude column name
  lngKey: string;       // Longitude column name
  coordFormat: 'DD' | 'DDM' | 'DMS'; // Coordinate format
  data: Array<RowObject>; // Row data with optional per-point overrides
}
```

5.2 Row Object (with Per-Point Overrides)

```
interface RowObject {
  [originalKey: string]: string | number | boolean | undefined;

  // Per-point overrides (not original data – injected at runtime):
  _showLabel?: boolean; // Show label for this point (overrides file.showLabels)
  _icon?: string | null; // Emoji icon override (overrides file.icon)
  _label?: string;       // Custom label text (checked first by getLabelText)
}
```

5.3 BUILTIN_SETS — Sidebar POI Definitions (Global Array, Read-Only)

```
interface BuiltinSet {
  id: string;          // 'poi_embassies' | 'poi_banks' | 'poi_government' |
                        'poi_pl_embassies'
  name: string;        // Display name in sidebar
  icon: string;        // Emoji character
  status: string;      // '' (not loaded) | 'loading' | 'N points loaded'
}
```

5.4 datasetMarkers — Marker References for Blink Animation

```
Record<string, L.Marker[]> // Indexed by fileId
```

Each file's Leaflet marker references are stored here so `blinkDataset()` can iterate and animate them. Markers are also inside `L.LayerGroup` instances on the map for rendering.

5.5 measurePoints — Measurement Waypoints

```
Array<{ lat: number; lng: number }> // Ordered list of click points
```

5.6 pendingImport — Temporary Import Buffer

```
interface PendingImport {
  rows: Array<Object>; // Parsed rows from SheetJS
  keys: Array<string>; // Column names
  name: string;        // Original filename
  type: 'xlsx' | 'csv'; // File type
  size: string;        // Formatted file size
}
```

Holds data between file pick and import confirmation. Cleared on cancel or after successful import.

5.7 Global State Variables

Variable	Type	Initial	Description
FILES	Array<FileEntry>	3 sample entries	All datasets (imported + generated)
currentFileId	string null	'f1'	Currently selected file ID
fileIdCounter	number	4	Auto-incrementing ID counter
sortKey	string null	null	Current sort column
sortDir	'asc' 'desc'	'asc'	Sort direction
mapMode	'points' 'track' 'both'	'points'	Map rendering mode

mapInstance	L.Map null	null	Leaflet map (lazy initialized)
tileLayer	L.TileLayer null	null	Leaflet tile layer (for theme swap)
measuring	boolean	false	Measure tool active state
measurePoints	Array<{lat, lng}>	[]	Measurement waypoints
pendingImport	PendingImport null	null	Data awaiting import confirmation
sidebarTab	'my' 'builtin'	'my'	Active sidebar tab
iconPickerTargetId	string null	null	File ID for open icon picker
removeTargetId	string null	null	File ID pending removal
renameTargetId	string null	null	File ID pending rename
currentDetailRow	object null	null	Row displayed in details panel
datasetMarkers	Record<string, L.Marker[]>	{}	Marker refs for blink
BUILTIN_SETS	BuiltinSet[]	4 entries	Read-only POI definitions
ICON_PICKER_OPTIONS	string[]	20 emoji	Available icon choices
DATASET_COLORS	string[]	10 hex colors	Cycled for dataset markers

6. UI Components

6.1 Static HTML Elements by ID

ID	Element Type	Role	Dynamic Content
#themeSwitch	<input type="checkbox">	Theme toggle	—
#recordCount		Header total record count	updateCounts()
#fileInput	<input type="file">	Hidden file picker (accepts .xlsx, .xls, .csv)	—
#fileList	<div class="file-list">	My Sets container	renderSidebar()
#builtinList	<div class="file-list">	Built-in POI container	renderBuiltinSets()
#tableCount		Table tab badge	updateCounts()

#mapCount		Map tab badge	updateCounts()
#paneTable	<div class="content-pane">	Table view pane	switchTab() toggles
#paneMap	<div class="content-pane">	Map view pane	switchTab() toggles
#currentFileLabel	<h3>	Active file name	renderTableBody()
#globalFilter	<input type="text">	Global search input	—
#clearFilter	<button class="clear-btn">	Clear search ×	.visible class toggled
#rowInfo		"X of Y rows" text	renderTableBody()
#dataTable	<table>	Main data table	renderTable()
#tableHead	<thead>	Table header row	renderTableHead()
#tableBody	<tbody>	Table body	renderTableBody()
#map	<div>	Leaflet container	renderMap()
#measureBtn	<button>	Measure toggle	.active class toggled
#clearMeasureBtn	<button>	Clear measurements	.hidden toggled
#totalDistDisplay		Distance total container	.hidden toggled
#totalDistVal		Distance value	renderMeasurements()
#detailsPanel	<div class="details-panel">	Slide-in panel	.open class toggled
#detailTitle	<h3>	Panel title	openDetails()
#detailBody	<div class="panel-body">	Panel content	openDetails()
#pointIconPicker	<div class="point-icon-picker">	Inline icon grid	openDetails() / togglePointIcon()

6.2 Modal Dialogs

ID	Width	State Variable	Confirm	Cancel	Purpose
#importDialog	680px	pendingImport	confirmImport()	cancelImport()	Column mapping formal select preview

#mergeDialog	460px	—	confirmMerge()	cancelMerge()	File select check
#removeDialog	400px	removeTargetId	confirmRemove()	cancelRemove()	Delete confir
#renameDialog	400px	renameTargetId	confirmRename()	cancelRename()	Inline editing
#iconPickerDialog	420px	iconPickerTargetId	selectIcon(ic)	cancelIconPicker()	Emoji select
#addPointDialog	480px	(pending)	confirmAddPoint()	cancelAddPoint()	Dynam field fo (SOON)

6.3 CSS Class Architecture

Class	Role	Key Properties
.app	Root flex column	height: 100vh, display: flex, flex-direction: column
.header	Top bar	height: 48px, display: flex, align-items: center, bordered
.sidebar	Left panel	width: 280px, display: flex, flex-direction: column, bordered
.file-item	Dataset row	display: flex, align-items: center, padding: 10px 12px, hover effects
.file-item.active	Selected state	border-left: 3px solid var(--accent)
.details-panel	Slide-in panel	position: absolute, right: calc(-1 * var(--panel-w)), transition: right 0.3s cubic-bezier(0.4, 0, 0.2, 1)
.details-panel.open	Visible state	right: 0
.modal-overlay	Dialog backdrop	position: fixed, inset: 0, background: rgba(0,0,0,0.6), z-index: 2000, flex centering
.modal	Dialog card	background: var(--surface), border-radius: 12px, box-shadow: 0 16px 48px rgba(0,0,0,0.4)
.marker-icon	Leaflet icon container	background: none, border: none, text-align: center
.marker-emoji	Emoji display	font-size: 24px, filter: drop-shadow(0 1px 2px rgba(0,0,0,0.5))

7. CSS Design System

7.1 Custom Properties

```
:root {
  --header-h: 48px;
  --sidebar-w: 280px;
  --panel-w: 340px;
  --bg: #0f1117;
  --surface: #1a1d27;
  --surface2: #232733;
  --border: #2a2e3a;
  --text: #e1e4eb;
  --text2: #8b90a0;
  --accent: #4f8cff;
  --accent2: #6c5ce7;
  --green: #00b894;
}
```

7.2 Dark Theme (Default)

Variable	Value	Usage
--bg	#0f1117	Page background
--surface	#1a1d27	Cards, sidebar, panels
--surface2	#232733	Secondary surfaces, hover states
--border	#2a2e3a	Borders, dividers
--text	#e1e4eb	Primary text
--text2	#8b90a0	Muted/secondary text

7.3 Light Theme (`.light` on `<body>`)

Variable	Value	Usage
--bg	#f0f2f5	Page background
--surface	#ffffff	Cards, sidebar, panels
--surface2	#e8eaeef	Secondary surfaces, hover states
--border	#d1d5db	Borders, dividers
--text	#1a1d27	Primary text
--text2	#6b7280	Muted/secondary text

7.4 Key Design Patterns

Pattern	Implementation
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Layout	Full-viewport flexbox column (.app height: 100vh) with header + body flex row
Sidebar	Fixed 280px, scrollable .file-list with overflow-y: auto, tabbed navigation
Content area	.center flex column with tab bar + .content (position: relative, flex: 1)
Details panel	Absolute-positioned, slides via right: calc(-1 * 340px) → right: 0, 0.3s cubic-bezier
Modals	Fixed overlay rgba(0,0,0,0.6), centered .modal card (400–680px), 12px radius
Scrollbars	Custom 6px Webkit scrollbar: transparent track, var(--border) thumb
Hover reveals	File item action buttons start at opacity: 0, transition to opacity: 1 on .file-item:hover
Segmented toggle	Toggle groups with overflow: hidden + border-radius: 8px for border collapse
Icon grids	CSS Grid: repeat(5, 1fr) in modal dialogs, repeat(6, 1fr) in details panel
Sort arrows	Inline span.sort-arrow with .active class for accent coloring
Leaflet overrides	!important on tooltip/popup backgrounds, borders to match theme
Transition	Blink uses CSS transition: filter 1s ease-in-out (sepia + blur → normal)

8. Leaflet Integration

8.1 Map Setup

Property	Value
Element	<div id="map"> (flex: 1 in .map-view)
Initialization	Lazy: first renderMap() call creates L.map('map', { zoomControl: false })
Default center	[39.8283, -98.5795] (geographic center of USA)
Default zoom	4
Zoom control	L.control.zoom({ position: 'bottomright' })

8.2 Tile Layers

Theme	URL
Dark	https://{s}.basemaps.cartocdn.com/dark_all/{z}/{x}/{y}{r}.png
Light	https://{s}.basemaps.cartocdn.com/light_all/{z}/{x}/{y}{r}.png
Swap method	tileLayer.setUrl(newUrl) — no layer re-creation

8.3 Layer Management

Layer Type	Grouping	Lifecycle
Dataset points	L.layerGroup per dataset	Removed by mapInstance.eachLayer() before re-render
Dataset tracks	L.polyline + start/end markers	Removed by mapInstance.eachLayer() before re-render
Measure points	Individual L.marker with _isMeasure = true	Removed by clearMeasurements() or map cleanup
Measure lines	Individual L.polyline with _isMeasure = true	Removed by clearMeasurements() or map cleanup
Measure labels	Individual L.marker with _isMeasure = true	Removed by clearMeasurements() or map cleanup
Dataset markers	Stored in datasetMarkers[fileId] array	References kept for blinkDataset()

8.4 Point Rendering

Type	Leaflet Component	When Used
Circle marker	L.circleMarker([lat, lng], { radius: 7, color: datasetColor, fillColor: datasetColor, fillOpacity: 0.8, weight: 2 })	Default — no icon override
Emoji icon	L.marker([lat, lng], { icon: L.divIcon({ className: 'marker-icon', html: '<div class="marker-emoji">EMOJI</div>', iconSize: [30, 30], iconAnchor: [15, 15] }) })	When file.icon or row._icon is set
Tooltip (label)	marker.bindTooltip(labelText, { permanent: true, direction: 'top', offset: [0, -14 or -18], className: 'point-label' })	When file.showLabels or row._showLabel is true
Popup	marker.bindPopup(html) — all non-lat/lng keys as key:value	On click
Click handler	marker.on('click', () => openDetails(row))	Always attached

8.5 Track Rendering

Element	Component	Style
Polyline	L.polyline(coords, options)	color: datasetColor, weight: 3, opacity: 0.8, smoothFactor: 1
Start marker	L.circleMarker(firstCoord, options)	radius: 8, fillColor: '#00b894' (green), color: '#fff'
End marker	L.circleMarker(lastCoord, options)	radius: 8, fillColor: '#e17055' (red), color: '#fff'

8.6 Measurement Tool

Element	Component	Style
Waypoints	L.marker with divIcon	Gold circle (14px), border: 3px solid #fdbc6e, draggable
Drag state	Class .measure-point-icon-dragging	Bigger (20px), orange tint
Segments	L.polyline	color: '#fdbc6e', weight: 2, dashArray: '6, 8'
Distance labels	L.marker at segment midpoint	Gold border, background: #2d2d2d, color: #fdbc6e
Total display	HTML #totalDistVal	Right-aligned, gold text
Distance calc	L.latLng(a).distanceTo(b)	Leaflet's built-in Haversine

8.7 Fit Bounds

```
mapInstance.fitBounds(L.latLngBounds(bounds), { padding: [30, 30] });
// bounds = all coords from focus file, or all visible files' coords
```

8.8 Blink Animation

```
// CSS transition on the marker element
transition: filter 1s ease-in-out;

// Apply blink
marker._icon.style.filter = 'sepia(1) blur(0.5px) hue-rotate(-30deg)';
// After 1s, remove - CSS transitions back over 1s
setTimeout(() => { marker._icon.style.filter = ''; }, 1000);
```

Each of the dataset's markers blinks sequentially with 100ms stagger between them for a cascading wave effect.

9. Scenario Flows

9.1 Import a CSV/XLSX File

#	Actor Action	System Response	Function Called
1	Clicks + Import button	Hidden <input #fileInput> clicked programmatically	importFile()
2	Selects .xlsx / .xls / .csv	File begins reading via FileReader	handleFileImport(event)
3	—	SheetJS parses data: XLSX.read() for xlsx, TextDecoder + XLSX.read() for csv	SheetJS API

4	—	Import dialog appears: column mapping selects, preview table, row count	<code>showImportDialog()</code>
5	—	Lat/lng columns auto-detected, coordinate format auto-detected	<code>getLatLngColumns(), detectCoordFormat()</code>
6	Adjusts lat/lng columns or format (optional)	Preview updates	<code>updateImportFormat()</code>
7	Clicks Import	Coordinates parsed, new FILES entry created, all views re-rendered	<code>confirmImport()</code>
8	—	Import dialog closes; data visible in table, map, and sidebar	Render chain
9	Clicks Cancel or ×	<code>pendingImport</code> cleared, modal dismissed	<code>cancelImport()</code>

9.2 Load a Built-in POI Set

#	Actor Action	System Response	Function Called
1	Clicks Built-in sidebar tab	Built-in list shown	<code>switchSidebarTab('builtin')</code>
2	Clicks e.g. "Embassies in Poland"	Item shows "loading..." state, disabled style	<code>loadBuiltinSet('poi_pl_embassies')</code>
3	—	Synthetic data generated (70 Warsaw embassy points)	<code>generatePolandEmbassiesData()</code>
4	—	All existing files deactivated	Loop: <code>f.active = false</code>
5	—	New FILES entry created (type: 'csv', cols: 5, auto-icon)	File constructor
6	—	Sidebar switches to "My Sets" tab	<code>switchSidebarTab('my')</code>
7	—	Built-in list updated with "70 points loaded"	<code>renderBuiltinSets()</code>
8	—	Full re-render: sidebar, table, map, counts	Render chain
9	—	Map centers on Warsaw, markers blink wave	<code>renderMap(), blinkDataset()</code>

9.3 Explore Points on Map

#	Actor Action	System Response	Function Called
---	--------------	-----------------	-----------------

1	Clicks a dataset in sidebar	Active file set, map zooms to bounds	<code>selectFile(id) → renderMap(id)</code>
2	Hovers over point marker	Tooltip appears (if labels enabled for dataset or point)	Leaflet <code>bindTooltip()</code>
3	Clicks point marker	Details panel slides in from right edge	<code>openDetails(row)</code>
4	—	Panel shows point title, toolbar (Label + Icon), key/value rows	Panel render
5	Clicks  Label button	<code>row._showLabel</code> toggled, panel + map re-rendered	<code>togglePointLabel()</code>
6	Clicks  Icon button	Inline emoji grid shown/hidden in panel	<code>togglePointIcon()</code>
7	Selects an emoji from grid	<code>row._icon</code> set, panel + map re-rendered with new icon	<code>selectPointIcon('🏠')</code>
8	Clicks × close button	Panel slides away, <code>currentDetailRow</code> cleared	<code>closeDetails()</code>

9.4 Sort and Filter Table

#	Actor Action	System Response	Function Called
1	Clicks "Revenue" column header	Table sorted ascending by revenue; ▲ shown	<code>sortBy('revenue')</code>
2	Clicks "Revenue" again	Sort direction toggles to descending; ▼ shown	<code>sortBy('revenue')</code>
3	Types "Paris" in global search	Table filters to rows where any column matches "Paris"; "X of Y" updated	<code>onFilterChange()</code>
4	Types ">1000" in Revenue column filter	Per-column filter applied, combined with global; further narrowed	<code>onFilterChange()</code>
5	Clicks × clear button	Global search cleared, per-column filters remain	<code>clearGlobalFilter()</code>
6	Clicks a table row	Details panel opens, map centers on point at zoom 10	<code>selectTableRow(idx)</code>

9.5 Use Measurement Tool

#	Actor Action	System Response	Function Called
1	Clicks measure button in map toolbar	Button active (<code>.active</code> class), cursor changes to crosshair	<code>toggleMeasure()</code>
2	Clicks a location on map	Gold circle placed; clear button appears	<code>addMeasurePoint(e)</code>

3	Clicks second location	Dashed line drawn between points; distance label at midpoint	<code>renderMeasurements()</code>
4	Clicks more points	Accumulating polyline with segment distances; total distance updated	<code>renderMeasurements()</code>
5	Drags a waypoint	Line + all distances update in real-time as marker moves	Drag handlers
6	Clicks $\times$ Clear	All measurements removed, UI reset	<code>clearMeasurements()</code>

9.6 Merge Two Datasets

#	Actor Action	System Response	Function Called
1	Clicks Merge button in sidebar header	Merge dialog opens with checkbox list of all datasets	<code>showMergeDialog()</code>
2	Checks ≥ 2 datasets	Selection state tracked	Checkbox change
3	Clicks Merge	Validates ≥ 2 checked; union of all keys computed	<code>confirmMerge()</code>
4	—	For each row: missing keys filled with ''	Data merge loop
5	—	New FILES entry created with name "Merged Dataset (N files)"	File constructor
6	—	All original files deactivated; new file active	State update
7	—	Sidebar, table, map, counts re-rendered	Render chain
8	—	Merge dialog closes	Modal close

9.7 Toggle Dark/Light Theme

#	Actor Action	System Response	Function Called
1	Clicks 🌙/☀️ toggle in header	Checkbox state toggles	<code>toggleTheme()</code>
2	—	<code>.light</code> class toggled on <code><body></code>	<code>classList.toggle('light')</code>
3	—	All CSS custom properties swap via cascade	CSS engine
4	—	Map tile layer URL swapped: <code>dark_all</code> $\leftrightarrow$ <code>light_all</code>	<code>tileLayer.setUrl()</code>
5	—	Full visual theme transition	CSS / tile load

9.8 Customize Dataset Icon

#	Actor Action	System Response	Function Called
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1	Clicks 🌈 on a file item (sidebar)	Icon picker modal opens with 20 emoji options	<code>showIconPicker(fileId)</code>
2	Hovers over emoji options	Visual hover highlight on each emoji cell	CSS <code>:hover</code>
3	Selects an emoji	<code>file.icon = selectedEmoji</code> , modal closes	<code>selectIcon('🏠')</code>
4	—	Sidebar re-rendered (icon shown in file item)	<code>renderSidebar()</code>
5	—	Map re-rendered: all points now use emoji markers	<code>renderMap()</code>
6	Clicks "Clear icon" (or × option)	<code>file.icon = null</code> , reverts to default circle markers	<code>selectIcon(null)</code>

9.9 Rename a Dataset

#	Actor Action	System Response	Function Called
1	Clicks 🖊 on a file item	Rename dialog opens with current name in input (auto-focused + selected)	<code>showRenameDialog(id)</code>
2	Types new name	Input value changes	—
3	Presses Enter or clicks Rename	Validates non-empty, updates <code>file.name</code> , re-renders sidebar + table	<code>confirmRename()</code>
4	Clicks Cancel or ×	Input discarded, state cleared	<code>cancelRename()</code>

9.10 Remove a Dataset

#	Actor Action	System Response	Function Called
1	Clicks × on a file item	Remove confirmation dialog opens with file name	<code>showRemoveDialog(id)</code>
2	Clicks Remove (red button)	File removed from FILES array	<code>confirmRemove()</code>
3	—	If active file removed: nearest neighbor selected (or null)	State cleanup
4	—	Sidebar, table, map, counts re-rendered	Render chain
5	Clicks Cancel or ×	<code>removeTargetId</code> cleared, dialog dismissed	<code>cancelRemove()</code>

9.11 Customize Per-Point Label (👉 SOON Planned)

#	Actor Action	System Response	Function Called
1	Opens details panel for a point	Label text input shown in toolbar area	<code>openDetails(row)</code>

2	Current label text displayed (auto-extracted or <code>_label</code>)	Input pre-filled	<code>getLabelText(row)</code>
3	Types new label text	Input updates	<code>onLabelInput()</code>
4	—	Map marker label updates in real-time	<code>reapplyLabel()</code>
5	Label persists when re-opening panel	<code>row._label</code> stored on row object	State persistence

9.12 Add New Point via Map Click (👉_{SOON} Planned)

#	Actor Action	System Response	Function Called
1	Clicks + Point in sidebar header	Add Point dialog opens	<code>showAddPointDialog()</code>
2	—	Dynamic form fields generated from active dataset schema	Form generation
3	Clicks 📍 button next to lat field	Map enters crosshair selection mode, cursor changes	<code>enableMapPick()</code>
4	Clicks a location on map	Lat + lng fields auto-populated, picker mode ends	<code>onMapPick(e)</code>
5	Fills remaining fields (optional)	Form data ready	User input
6	Clicks Add Point	New row appended at end of <code>file.data</code>	<code>confirmAddPoint()</code>
7	—	Table, map, sidebar re-rendered	Render chain
8	—	New point visible on map with dataset's icon/color	<code>renderMap()</code>

10. Implementation Status

10.1 Legend

Icon	Meaning
✅	Completed and tested
👉 _{SOON}	Planned, approved, not yet implemented
❌	Not planned / future consideration

10.2 Feature Completeness Matrix

Category	Feature	ID	Status
Data	Import CSV/XLSX	F1	✅

Data	Coordinate auto-detection	F2	✓
Data	Multi-format coordinate parsing	F3	✓
Data	Merge datasets	F4	✓
Data	Remove dataset	F5	✓
Data	Rename dataset	F6	✓
Data	Dataset visibility toggle	F7	✓
Data	Built-in POI sets	F8	✓
Data	Add new point	F9	→ SOON
Data	Dataset-level icon	F10	✓
Map	Point markers	M1	✓
Map	Per-point icon override	M2	✓
Map	Track mode	M3	✓
Map	Combined mode	M4	✓
Map	Dataset labels	M5	✓
Map	Per-point labels	M6	✓
Map	Auto label extraction	M7	✓
Map	Per-point custom label	M8	→ SOON
Map	Map click → details	M9	✓
Map	Zoom to dataset	M10	✓
Map	Blink animation	M11	✓
Map	Measurement tool	M12	✓
Map	Theme-aware tiles	M13	✓
Map	Map click coordinate pick	M14	→ SOON
Map	Popup on click	M15	✓
Table	Column sorting	T1	✓
Table	Global text search	T2	✓
Table	Per-column filters	T3	✓
Table	Row selection → details	T4	✓
Table	Row count display	T5	✓
Table	Clear filter button	T6	✓

Details	Slide-in panel	D1	✓
Details	Key/value display	D2	✓
Details	Per-point label toggle	D3	✓
Details	Per-point icon picker	D4	✓
Details	Per-point label editing	D5	➡ SOON
UX	Dark/Light theme	U1	✓
UX	Main tab switching	U2	✓
UX	Total record counter	U3	✓
UX	Sidebar tab system	U4	✓
UX	Loading indicators	U5	✓
UX	Empty states	U6	✓
UX	Measurement distance	U7	✓

10.3 Upcoming Work (Approved)

Feature	ID	Scope	Complexity
Per-point custom label editing	D5 / M8	Add text input in details panel below toolbar; <code>getLabelText()</code> checks <code>row._label</code> first; map updates in real-time	Low — ~20 lines
Add new point with map click	F9 / M14	+ <code>Point</code> button in sidebar header; modal with dynamic fields; map-click coordinate picker; confirm appends + re-renders	Medium — ~80 lines

Appendices

A. Coordinate Parsing Details

`parseCoord(value, format)` supports three formats:

Format	Example	Parsing Logic
DD (Decimal Degrees)	40.7128	<code>parseFloat()</code>
DDM (Degrees Decimal Minutes)	40°42.768' N	<code>deg + min/60</code> , apply sign from N/S/E/W
DMS (Degrees Minutes Seconds)	40°42'46" N	<code>deg + min/60 + sec/3600</code> , apply sign from N/S/E/W

`detectCoordFormat(values)` heuristic:

- 1. Check if values contain ° (degree symbol)
- 2. If yes: check for " (double prime) → 'DMS' ; check for ' (prime) → 'DDM'
- 3. Otherwise → 'DD'

B. Color Palette (DATASET_COLORS)

```
[ '#4f8cff', '#6c5ce7', '#00b894', '#fdcb6e', '#e17055',  
  '#00cec9', '#a29bfe', '#fd79a8', '#74b9ff', '#55efc4' ]
```

Cycled through modulo `i % 10` for up to 10 datasets before repeating.

C. Icon Picker Options (ICON_PICKER_OPTIONS)

```
[ '📍', '🏛️', '🏢', '🏠', '🏡', '🏢', '🏠', '🏡', '🏢', '🏠',  
  '🏡', '🏢', '🏠', '🏡', '🏢', '🏠', '🏡', '🏢', '🏠', '🏡',  
  '🏢', '🏠', '🏡', '🏢', '🏠', '🏡', '🏢', '🏠', '🏡', '🏢' ]
```

20 emoji choices in both dataset-level (modal, 5-column grid) and per-point (inline, 6-column grid) pickers.

D. Built-in Dataset Generators — Coverage

Generator	Cities	Range	Jitter	Dedup
<code>generateEmbassiesData()</code>	18 capitals	2–6 per city	$\pm 0.02^\circ$	No (countries unique per city)
<code>generateBanksData()</code>	10 hubs	3–7 per city	$\pm 0.015^\circ$	Yes (Set-based)
<code>generateGovernmentData()</code>	12 capitals	3–6 per city	$\pm 0.0125^\circ$	Yes (Set-based)
<code>generatePolandEmbassiesData()</code>	1 (Warsaw)	70 total	0.01–0.03° radial	No (all unique countries)

Document generated: May 2026
Source: `index.html` (~2243 lines)
Next planned features: Per-point label editing, Add new point with map-click coordinate picker